

Tarea Semanal 1

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Dado $h[n] = \delta[n] - \delta[n - 4]$, encontrar $y[n] = x[n] * h[n]$ para cada una de las siguientes $x[n]$:

a) $x[n] = \cos(\omega_0 \cdot n \cdot T_s)$

Expresar la respuesta como un único coseno de la forma $A \cos(\omega_0 \cdot n \cdot T_s + \varphi)$.

$$\begin{aligned} y[n] &= x[n] * (\delta[n] - \delta[n - 4]) \\ &= x[n] * \delta[n] - x[n] * \delta[n - 4] \\ &= x[n] - x[n - 4] \end{aligned} \quad (1)$$

$$x[n] - x[n - 4] = \cos(\omega_0 \cdot n \cdot T_s) - \cos(\omega_0 \cdot (n - 4) \cdot T_s) = \cos(\omega_0 \cdot n \cdot T_s) - \cos(\omega_0 \cdot n \cdot T_s - \omega_0 \cdot 4 \cdot T_s) \quad (2)$$

Defino $\theta = \omega_0 \cdot n \cdot T_s$ y $\varphi = \omega_0 \cdot 4 \cdot T_s$.

Además, uso la identidad trigonométrica:

$$\cos A - \cos B = -2 \sin\left(\frac{A + B}{2}\right) \sin\left(\frac{A - B}{2}\right) \quad (3)$$

Luego,

$$\begin{aligned} \cos \theta - \cos(\theta - \varphi) &= -2 \sin\left(\frac{\theta + \theta - \varphi}{2}\right) \sin\left(\frac{\theta - \theta + \varphi}{2}\right) \\ &= -2 \sin\left(\frac{2\theta - \varphi}{2}\right) \sin\left(\frac{\varphi}{2}\right) \\ &= -2 \sin\left(\theta - \frac{\varphi}{2}\right) \sin\left(\frac{\varphi}{2}\right) \end{aligned} \quad (4)$$

Sabiendo que $\sin \theta = \cos\left(\theta - \frac{\pi}{2}\right)$:

$$= \underbrace{-2 \sin\left(\frac{\varphi}{2}\right)}_A \cos\left(\theta - \frac{\varphi}{2} - \frac{\pi}{2}\right) \quad (5)$$

$$= A \cdot \cos\left(\omega_0 \cdot n \cdot T_s - \omega_0 \cdot 4 \cdot T_s - \frac{\pi}{2}\right) = A \cos(\omega_0 \cdot n \cdot T_s + \varphi_1) \quad (6)$$

Donde:

- $A = -2 \sin\left(\frac{\omega_0 \cdot 4 \cdot T_s}{2}\right) = -2 \sin(2\omega_0 T_s)$
- $\varphi_1 = -4\omega_0 T_s - \frac{\pi}{2}$

b) $x[n] = \left(\frac{1}{2}\right)^n \cdot u[n]$

$$x[n-4] = \left(\frac{1}{2}\right)^{n-4} \cdot u[n-4] \quad (7)$$

Donde:

$$u[n-a] = \begin{cases} 0 & n < a \\ 1 & n \geq a \end{cases} \quad (8)$$

$$y[n] = x[n] - x[n-4] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4] \quad (9)$$

Analizando por intervalos:

$$y[n] = \begin{cases} 0 & n < 0 \\ \left(\frac{1}{2}\right)^n & 0 \leq n < 4 \\ \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4} & n \geq 4 \end{cases} \quad (10)$$

Para $n \geq 4$:

$$\begin{aligned} \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4} &= \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^{-4} \\ &= \left(\frac{1}{2}\right)^n - 16 \cdot \left(\frac{1}{2}\right)^n \\ &= -15 \cdot \left(\frac{1}{2}\right)^n \end{aligned} \quad (11)$$

Por lo tanto:

$$y[n] = \begin{cases} 0 & n < 0 \\ \left(\frac{1}{2}\right)^n & 0 \leq n < 4 \\ -15 \cdot \left(\frac{1}{2}\right)^n & n \geq 4 \end{cases} \quad (12)$$

c) $x[n] = u[n+1] - u[n-2]$

$$x[n-4] = u[n-3] - u[n-6] \quad (13)$$

$$y[n] = \underbrace{u[n+1]}_{1 \text{ si } n \geq -1} - \underbrace{u[n-2]}_{1 \text{ si } n \geq 2} - \underbrace{u[n-3]}_{1 \text{ si } n \geq 3} + \underbrace{u[n-6]}_{1 \text{ si } n \geq 6} \quad (14)$$

Analizando los intervalos:

$$y[n] = \begin{cases} 0 & n < -1 \\ 1 & -1 \leq n < 2 \\ 0 & 2 \leq n < 3 \\ -1 & 3 \leq n < 6 \\ 0 & n \geq 6 \end{cases} \quad (15)$$